

**A role-aware explainable AI dashboard for at-risk student  
classification: Design and technical evaluation on the Open  
University Learning Analytics Dataset**

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Running title: Role-aware explainable AI dashboard for at-risk students

## 10 **Highlights**

- 11 • One prediction system serves instructors, advisors and administrators separately.
- 12 • XGBoost reached 0.909 recall and 0.887 F1 with course presentations held out.
- 13 • SHAP and LIME shared 7 of 10 top features but only 0.635 top-5 overlap per student.
- 14 • Accuracy varied by over seven points across the four held-out presentations.
- 15 • CommonKADS traces each dashboard view to a stated role-specific knowledge need.
- 16

## 17 **Abstract**

18 Predictive models built on learning management system (LMS) data can flag students at risk  
19 of failing or withdrawing, but most systems present isolated risk scores without the reasoning  
20 behind them and give every staff role the same view. This study designed, implemented and  
21 technically evaluated a role-aware explainable artificial intelligence (XAI) dashboard that  
22 delivers explanations from one prediction system to instructors, academic advisors and  
23 administrators at different levels of aggregation. Following design science research with the  
24 CommonKADS knowledge engineering methodology, Random Forest, XGBoost and  
25 Gradient Boosting classifiers were evaluated on the Open University Learning Analytics  
26 Dataset (32,593 student-module-presentation observations) using four-fold cross-validation  
27 grouped by course presentation. SHAP and LIME formed the explanation layer, and the  
28 prototype was integrated with Moodle and Canvas. The XGBoost prototype, selected on  
29 recall, achieved out-of-fold accuracy of 0.877, precision of 0.866, recall of 0.909, F1-score of  
30 0.887 and AUC of 0.962; Random Forest obtained a marginally higher F1-score (0.898).  
31 Accuracy ranged from 0.846 to 0.916 across held-out presentations. Both explanation  
32 methods ranked the number of assessments submitted first and shared seven of their top ten  
33 features, but the mean instance-level top-5 overlap was only 0.635. A role-aware explanation  
34 layer is technically feasible at full dataset scale, and the partial agreement between SHAP and  
35 LIME supports showing both methods side by side. The features cover the full presentation,  
36 so results describe retrospective classification; practitioner trust, calibration and fairness  
37 remain to be evaluated.

38 **Keywords:** Explainable artificial intelligence; Learning analytics dashboards; At-risk student  
39 prediction; SHAP; LIME; Knowledge engineering; OULAD

40

## 1. Introduction

Learning management systems (LMS) such as Moodle, Canvas and Blackboard now support course delivery, assessment and communication at scale, and their use grew substantially when teaching moved online during the COVID-19 pandemic, increasing the volume of behavioural data available for analysis (Rebelo Marcolino et al., 2025). Learning analytics, defined as the measurement, collection, analysis and reporting of data about learners and their contexts to understand and optimise learning (Siemens & Long, 2011), has used these data to identify students at risk of failing or withdrawing early enough for staff to intervene (Jin et al., 2024; Kuzilek et al., 2017). The Open University Learning Analytics Dataset (OULAD), which records 32,593 student-module-presentation observations across 22 module presentations (Kuzilek et al., 2017), has become one of the most widely used benchmarks for this work (Gunasekara & Saarela, 2025; Jin et al., 2024; Wang, 2025).

Predictive accuracy on such benchmarks is now high, but less attention has been given to what happens after a student is flagged. Many prediction systems generate risk scores without explaining the reasoning behind them, leaving staff without enough context to decide whether and how to act (Susnjak et al., 2022). Explainable AI (XAI) techniques such as SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME) can attribute a prediction to the features that drove it, yet their integration into working learning analytics dashboards remains limited (Ahmed et al., 2025; Ali et al., 2023). A further problem is that institutional stakeholders need different kinds of explanation from the same model: an instructor monitoring a running course does not need the same view as an administrator reviewing withdrawal across a cohort (Kostopoulos et al., 2024). The literature suggests that explanations should reflect the recipient's role and expertise (Haque et al., 2023; Kostopoulos et al., 2024), but the reviewed studies did not identify a dashboard that delivers role-specific explanations to several staff roles from a single prediction system (Afzaal et al., 2021; Cabral et al., 2025; Kaliisa et al., 2023; Susnjak et al., 2022).

This study treats the problem as both a design problem and a technical one. It designs, implements and technically evaluates a role-aware XAI dashboard for at-risk student

classification, guided by the CommonKADS knowledge engineering methodology (Schreiber et al., 2000), and asks three research questions:

- RQ1. How can SHAP and LIME be integrated into a knowledge engineering pipeline to improve transparency in student at-risk prediction models?
- RQ2. What dashboard features and explanation structures, based on the CommonKADS Agent Model and the learning analytics dashboard literature, can be specified to support the different information needs of instructors, advisors and administrators within an LMS?
- RQ3. What do the predictive performance and explainability findings indicate about the technical potential of the role-aware dashboard to support practitioner interpretation and decision-making?

The study makes four contributions: (1) an implemented role-aware XAI dashboard architecture that differentiates SHAP and LIME explanations across instructor, advisor and administrator views of one LMS-integrated prediction system; (2) a documented ensemble model evaluated on OULAD under a protocol that holds out entire course presentations; (3) technical evidence on the agreement and divergence of SHAP and LIME on a real educational dataset; and (4) candidate design guidelines for explainable learning analytics dashboards, derived from the CommonKADS models and the implemented system. The evaluation is technical and uses secondary data only. Findings are therefore reported at three levels of evidence: properties demonstrated by measurement, properties supported by the design but not tested with users, and properties not empirically established.

## **2. Related work**

### *2.1. At-risk prediction and evaluation protocols*

Research on at-risk prediction spans logistic regression, decision trees, ensembles and deep learning. In a systematic review of OULAD studies, Jin et al. (2024) found that ensemble methods, particularly gradient boosting variants and random forests, are common and typically outperform simpler classifiers, while practitioner-facing aspects receive little attention. Reported accuracies can be very high: Wang (2025) reports 96.9% accuracy on

OULAD for a hybrid ensemble, and Rebelo Marcolino et al. (2025) report strong dropout prediction on Moodle logs using CatBoost with multi-objective optimisation. Guevara-Reyes et al. (2025) show that more complex ensembles gain accuracy at the cost of intrinsic interpretability. Engagement-based features tend to outperform static demographic ones (Hussain et al., 2018), useful predictions can be made after a fraction of a course has elapsed (Adnan et al., 2021), and deadline-anchored activity windows have been reported to give stronger early-warning signals on OULAD than calendar windows (Hlosta et al., 2017).

Reported figures depend strongly on the evaluation protocol. Jin et al. (2024) note that protocols vary considerably across OULAD studies, Krstajic et al. (2014) show that letting observations from the same higher-level unit appear in both training and validation folds produces optimistic estimates, and Pelánek (2018) argues that cross-validation, filtering and metric choices materially change reported performance in student modelling independently of model quality. Fairness is a further concern: Baker and Hawn (2022) document systematic performance disparities across demographic subgroups in educational models, and Yu et al. (2020) show that error rates can differ across student groups even when overall accuracy is high.

## 2.2. Explainable AI in education

XAI surveys distinguish intrinsically interpretable models from post-hoc explanation techniques and argue that the choice depends on the deployment context and the expertise of the user (Barredo Arrieta et al., 2020). Ali et al. (2023) report that tailoring explanations to user types remains an open problem, with most systems defaulting to technically framed outputs. In education, SHAP (Lundberg et al., 2020; Lundberg & Lee, 2017) and LIME (Ribeiro et al., 2016) dominate current applications. Nnadi et al. (2024) found SHAP the most consistent of several methods for attribution stability, Salih et al. (2025) discuss the computational cost of SHAP, its feature independence assumption and the sensitivity of LIME to sampling, and Gunasekara and Saarela (2025) apply both methods to models trained on OULAD. Hooshyar and Yang (2024) report significant variability in the feature importance assigned by SHAP variants and LIME to the same educational model. Critiques

of post-hoc explanation add caution: Rudin (2019) argues that high-stakes decisions should rest on intrinsically interpretable models, and Slack et al. (2020) show that both SHAP and LIME can be manipulated through their perturbation schemes.

Work on the human side of explanation argues that explanations should be contrastive, selective and suited to the recipient (Miller, 2019), and that their effectiveness depends on the fit between content, expertise and decision (Ehsan et al., 2021; Liao & Varshney, 2022). Khosravi et al. (2022) identify several stakeholder groups in educational XAI whose explanation needs differ, and Conati et al. (2021) show that learners with different prior knowledge derive different value from the same explanations. Afzaal et al. (2021) present an early example of SHAP-based explanations delivered within an LMS, but for students rather than for the staff who make intervention decisions.

### *2.3. Learning analytics dashboards and role-aware design*

Learning analytics dashboards (LADs) are the main channel through which analytics reach staff, but reviews report limited theoretical grounding and evaluation (Matcha et al., 2020; Schwendimann et al., 2017). Cabral et al. (2025) identify limited XAI integration as one of the most persistent gaps in AI-powered LADs. Existing designs typically serve one user group: Susnjak et al. (2022) combine descriptive, predictive and prescriptive analytics for learners, Chavan and Mitra (2022) design for instructors, and Kaliisa and Dolonen (2023) found that instructors prioritise summary-level alerts over granular breakdowns. Even within one role, needs vary; in the LADA advising dashboard, senior advisors preferred cohort views while less experienced advisors relied on individual detail (Gutiérrez et al., 2020). Kaliisa et al. (2023) provide a checklist for planning, designing, implementing and evaluating LADs that calls for engagement with the full range of stakeholders. Knowledge engineering offers a structured way to do this: CommonKADS models agents, their tasks and the knowledge they require as distinct artefacts (Schreiber et al., 2000; Studer et al., 1998). Taken together, the reviewed literature shows mature technical components for prediction and explanation, but no staff-facing design that serves instructors, advisors and administrators from one prediction system with role-specific explanations.

### 3. Method

#### 3.1. Research design

The study follows the Design Science Research Methodology (Hevner et al., 2004; Peffers et al., 2007), with CommonKADS (Schreiber et al., 2000) as the knowledge engineering and design framework applied within it. The Technology Acceptance Model (Davis, 1989) was used only as a design lens for perceived usefulness and ease of use, and the checklist of Kaliisa et al. (2023) guided dashboard design. Table 1 summarises how each of the six CommonKADS models was instantiated. Because no institutional partner or practitioners were involved, the Organisation and Agent Models were populated from published literature rather than from knowledge elicitation, which CommonKADS would ordinarily expect; the resulting role characterisations are therefore well-sourced hypotheses rather than documented accounts.

Table 1. Instantiation of the six CommonKADS models in this study.

| Model         | Instantiation                                                                                                                                                                                                                                             | Resulting design decision                                                                        |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Organisation  | How at-risk information reaches instructors, advisors and administrators and where the flow breaks down (scores without explanation; one view for all roles), from published accounts (Kaliisa et al., 2023; Leitner et al., 2019; Tsai & Gašević, 2017). | Three staff roles served from one shared prediction system; students out of scope.               |
| Task          | Risk assessment, explanation generation, role assignment and explanation delivery; intervention left to the human agent (Verbert et al., 2013).                                                                                                           | Layered architecture separating data, prediction, explanation, role assignment and presentation. |
| Agent         | Knowledge, capabilities and explanation needs of each role, from literature rather than elicitation (Haque et al., 2023; Kostopoulos et al., 2024).                                                                                                       | Three information briefs at individual, caseload and cohort granularity (Table 2).               |
| Knowledge     | Engagement, assessment and demographic features and their SHAP attributions (Kuzilek et al., 2017; Lundberg & Lee, 2017).                                                                                                                                 | Feature families with a human-readable feature dictionary.                                       |
| Communication | Selective explanations phrased for the recipient (Miller, 2019).                                                                                                                                                                                          | Progressive disclosure, side-by-side SHAP and LIME, role-gated views.                            |
| Design        | Mapping of the five models above onto the technical architecture (Schreiber et al., 2000).                                                                                                                                                                | Five-layer architecture with role-based access control (Section 4).                              |

### 3.2. Dataset and target

OULAD comprises seven relational tables covering registrations, demographics, module presentations, assessments, assessment results, VLE resources and VLE interaction logs (Kuzilek et al., 2017). These were joined into one row per student, module and presentation, giving 32,593 observations. Following the convention in the OULAD literature (Jin et al., 2024), Withdrawn ( $n = 10,156$ ) and Fail ( $n = 7,052$ ) outcomes were mapped to the at-risk class (17,208 observations, 52.8%) and Pass ( $n = 12,361$ ) and Distinction ( $n = 3,024$ ) to the not-at-risk class (15,385, 47.2%). The classes are only mildly imbalanced. Because withdrawals account for 59% of the positive class, the model is best understood as substantially a withdrawal predictor.

### 3.3. Features

The model input comprised 24 columns (17 numeric and 7 categorical), expanding to 59 features after one-hot encoding (Appendix B). The features form three families. The engagement family contains total VLE clicks, active days, mean daily clicks and clicks in 7-, 14- and 28-day windows before each assessment deadline, following the deadline-anchored strategy of Hlosta et al. (2017); a programmatic check raises an error if any window includes activity on or after its deadline. The assessment family contains the number of assessments submitted, mean and minimum score, lateness, late submissions and banked results. The demographic and registration family contains module, gender, region, highest education, index of multiple deprivation (IMD) band, age band, disability, previous attempts, studied credits, registration date and presentation length. Counts were set to zero where no activity was recorded; other missing values were imputed inside the model pipeline (median or most frequent value) using the training folds only. All behavioural and assessment features were computed over the full course presentation, so the evaluation measures retrospective classification rather than performance at a predefined early-warning point.

### 3.4. Models and evaluation protocol

Three ensembles representing different points on the bias-variance spectrum were compared: Random Forest (Breiman, 2001), XGBoost (Chen & Guestrin, 2016) and scikit-learn

195 Gradient Boosting (Friedman, 2001; Pedregosa et al., 2011). Evaluation used four-fold  
196 GroupKFold cross-validation grouped by course presentation (2013B, 2013J, 2014B and  
197 2014J), so no presentation appears in both the training and validation partitions of a fold,  
198 avoiding the leakage pattern described by Krstajic et al. (2014). The folds were not stratified,  
199 since stratification is incompatible with grouping on a variable with four values. XGBoost  
200 used a fixed configuration (300 estimators, maximum depth 5, learning rate 0.05, subsample  
201 0.8 and column subsample 0.8) without hyperparameter search or resampling, and the other  
202 two models were run under the identical preprocessing pipeline and protocol. Accuracy,  
203 precision, recall and F1-score were computed over out-of-fold predictions for all 32,593  
204 observations at a 0.5 threshold, together with ROC-AUC. F1-score was used for comparative  
205 reporting, while recall was the primary criterion for selecting the prototype model: a student  
206 who is at risk but not flagged receives no support, whereas a student flagged unnecessarily is  
207 reviewed by a practitioner and can be set aside.

### 208 3.5. *Explanation analysis*

209 Global importance was computed with TreeSHAP (Lundberg et al., 2020) using the final  
210 fitted model across all 32,593 observations, ranking the 59 transformed features by mean  
211 absolute SHAP value in log-odds units; these values explain the final model and are not fold-  
212 specific. LIME (Ribeiro et al., 2016) was run on a random sample of 40 observations  
213 (NumPy default\_rng, seed 42), with explanations aggregated by mean absolute weight.  
214 Agreement was assessed at two levels: overlap of the aggregate top-10 rankings, and the  
215 mean overlap between each sampled instance's top-5 SHAP and top-5 LIME features. To test  
216 whether the ordering of two near-duplicate engagement features was stable, the final model  
217 was refitted with seven further random seeds. The explanation stability and fidelity checks  
218 originally specified, following Visani et al. (2022), were not completed, so the explanation  
219 evaluation is limited to cross-method agreement.

## 220 4. System design and implementation

### 221 4.1. Architecture

222 The system has five layers (Fig. 1): a Data Layer for feature engineering, a Prediction Engine  
223 for risk classification, an Explainability Module for SHAP and LIME, an Interface/API layer  
224 enforcing authentication and role-based access control, and a Presentation Layer with three  
225 role-differentiated views. The layers communicate through REST endpoints, so each can be  
226 revised independently, and explanation is treated as a core architectural component rather  
227 than an addition to the model (Mohseni et al., 2021). The fitted preprocessing pipeline and  
228 model are saved as one versioned artefact with a manifest recording the training date, feature  
229 dictionary version and evaluation metrics, so any prediction returned by the API can be  
230 traced to the model version and feature contract that produced it. A versioned feature  
231 dictionary also carries a human-readable label for each feature, so explanations are shown  
232 with meaningful labels rather than variable names (Miller, 2019).

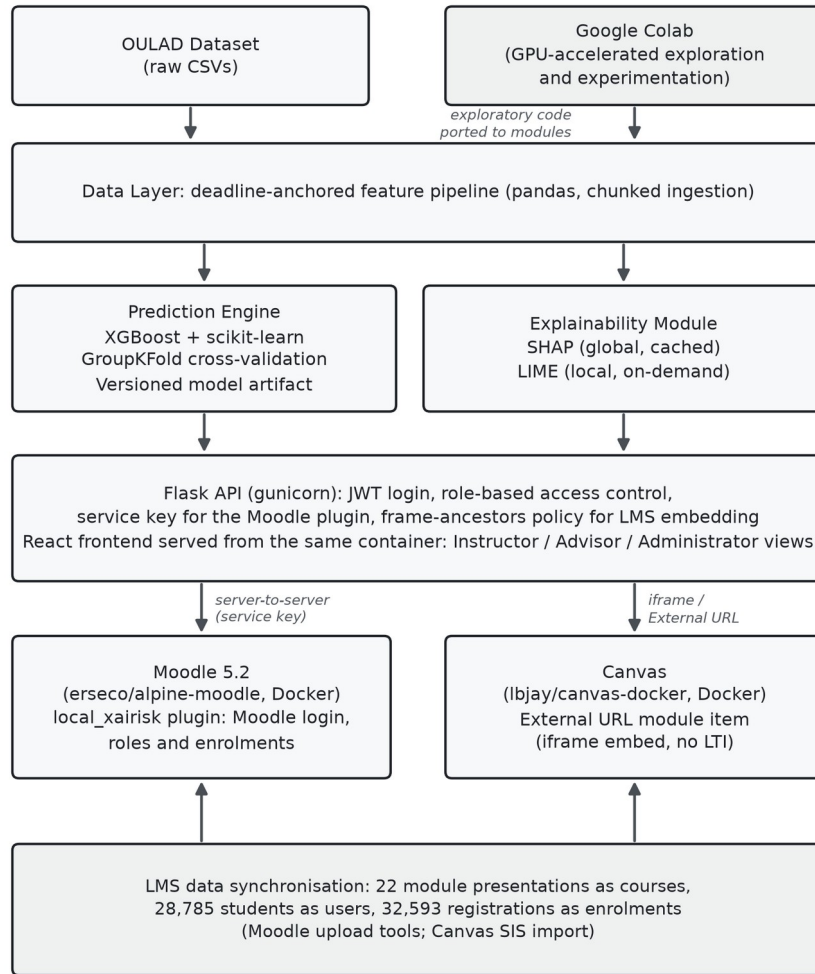
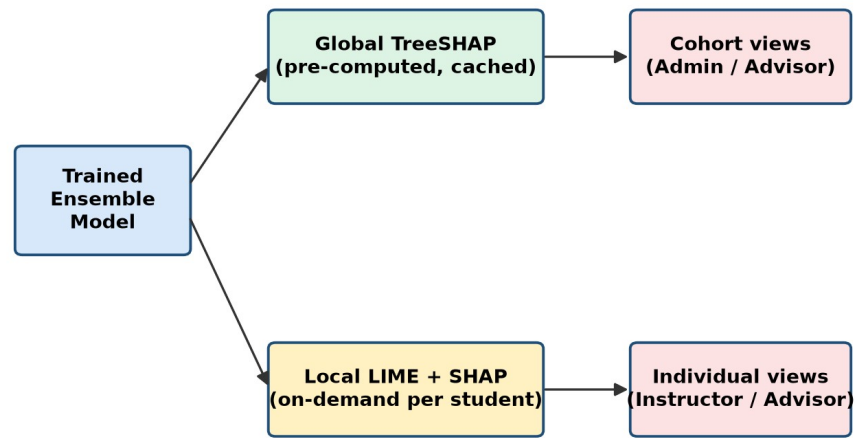


Fig. 1. Implemented system architecture, showing the data, prediction, explainability, API and presentation layers, the model development environment and the two LMS integration paths tested. All services run locally under Docker Compose.

#### 4.2. Explanation services

The Explainability Module has two services (Fig. 2). A global service computes TreeSHAP values once per model version and caches them for the cohort-level views. A local service computes a local SHAP explanation and an independent LIME explanation on demand for a single prediction, using the exact feature vector the model scored. Caching the global values reflects the higher computing time of SHAP reported by Salih et al. (2025), whereas LIME relies on perturbation and must be regenerated for each instance. Both explanations are shown side by side so that a user can see where they agree and where they differ, following the argument of Salih et al. (2025) for using the two methods together. Following the selectivity property of explanations (Miller, 2019) and evidence that comprehension degrades

247 when full attribution vectors are displayed (Poursabzi-Sangdeh et al., 2021), the interface  
 248 shows a small number of leading contributions rather than the complete vector.



249  
 250 Fig. 2. Dual-service explainability pipeline: global TreeSHAP values are pre-computed and cached  
 251 for cohort views, while LIME and local SHAP are generated on demand for individual students.

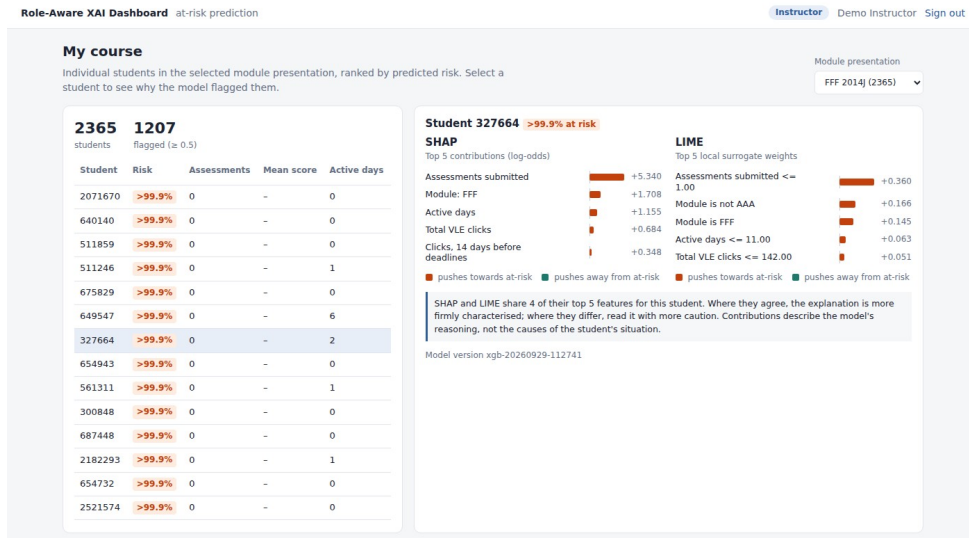
#### 252 4.3. Role-aware views

253 Each view was designed around an information brief derived from the Agent Model (Table  
 254 2). The instructor view gives per-student detail with SHAP and LIME shown side by side,  
 255 following evidence that teachers act on dashboard information at the level of the individual  
 256 learner (Molenaar & Knoop-van Campen, 2019). The advisor view averages the reasoning  
 257 across a caseload of students at or above the watch threshold, following the caseload  
 258 orientation of LADA (Gutiérrez et al., 2020). The administrator view shows institution-wide  
 259 model performance and a full-population reasoning trail, in line with institutional decision-  
 260 support patterns (Tsai & Gašević, 2017). All three views reuse one "reasoning trail"  
 261 component, a chart of signed SHAP and LIME contributions, at different levels of  
 262 aggregation, so the roles differ in the granularity rather than the visual form of the knowledge  
 263 they receive (Sokol & Flach, 2020). Progressive disclosure is applied throughout: compact  
 264 risk summaries appear by default and attribution detail on request (Springer & Whittaker,  
 265 2020). Fig. 3 shows the three views.

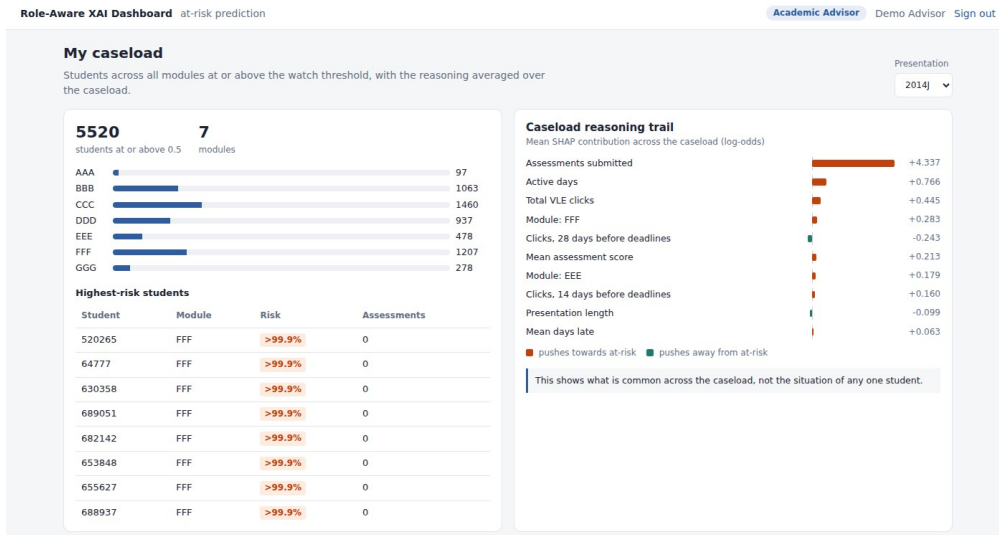
266 Table 2. Role-based dashboard design.

| Role          | Decision                                                        | Information needed                                                                                                         | Dashboard component                                                             | Granularity |
|---------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-------------|
| Instructor    | Whether to act within a course that is currently running.       | Class-level engagement in the current period and per-student risk drivers at the level of assessments and weekly activity. | Per-student reasoning trail with SHAP and LIME side by side.                    | Individual  |
| Advisor       | How to approach a particular student and prioritise a caseload. | Views across modules for a caseload, with demographic context suited to one-to-one consultations.                          | Caseload-averaged reasoning trail for students at or above the watch threshold. | Caseload    |
| Administrator | Where to direct institutional attention and resources.          | Cohort-level aggregations and module-over-module comparisons.                                                              | Institution-wide performance summary and full-population reasoning trail.       | Cohort      |

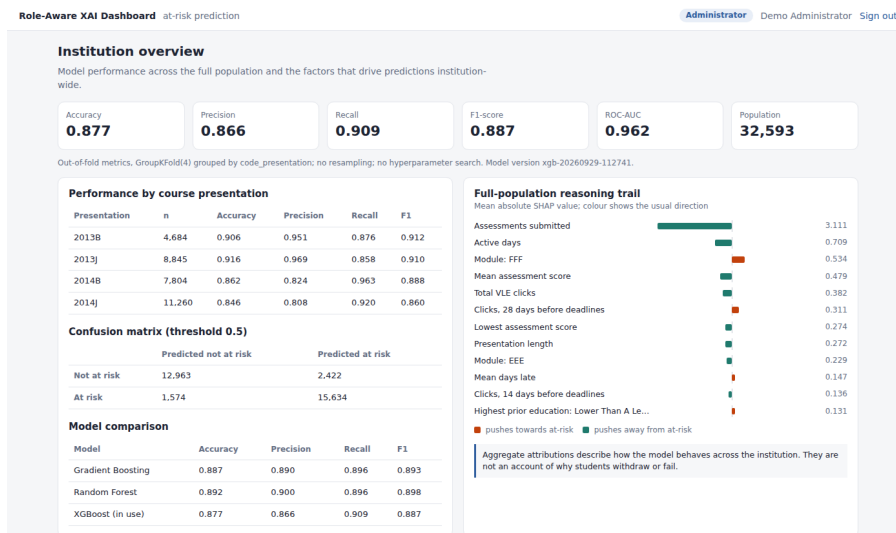
267



(a) Instructor view: per-student SHAP and LIME reasoning trail.



(b) Advisor view: caseload-averaged reasoning trail for students at or above the watch threshold.

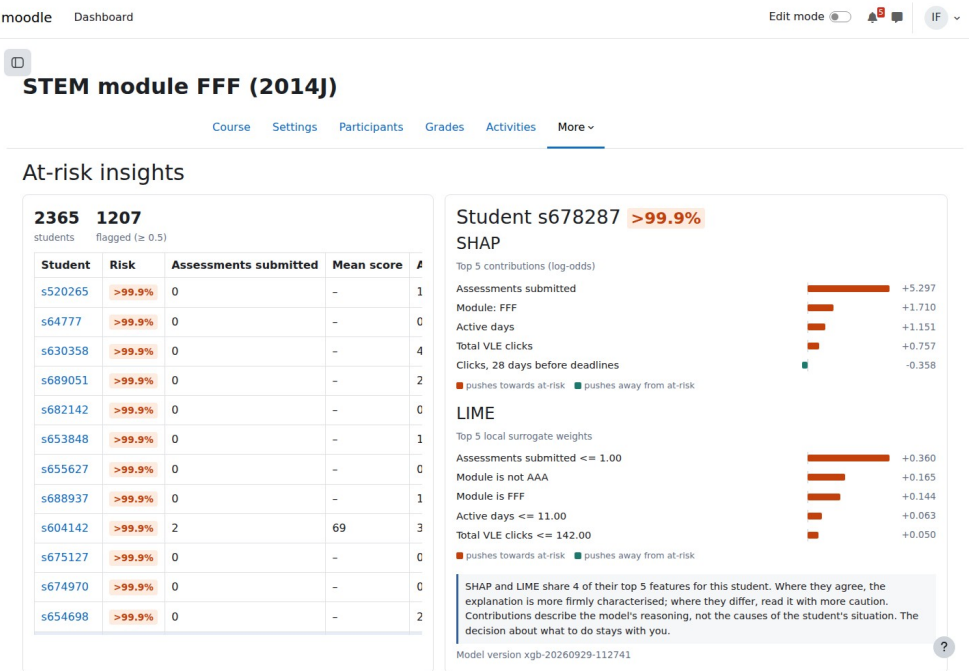


(c) Administrator view: institution-wide performance summary and full-population reasoning trail.

Fig. 3. The three role-gated dashboard views rendered from the same prediction and explanation services.

276 4.4. Security and LMS integration

277 Role separation is enforced at both the LMS and the API, following the defence-in-depth  
278 principle of Saltzer and Schroeder (1975). The API accepts either a JSON Web Token (Jones  
279 et al., 2015) from its own login or, for server-to-server calls from the Moodle plugin, a shared  
280 service key, and it checks the forwarded role against each endpoint. The visible view is gated  
281 by the authenticated session, with no client-side mechanism to switch roles. Two integration  
282 paths were tested locally in Docker. In Moodle 5.2 the dashboard was integrated as a native  
283 local plugin, with all 22 module presentations loaded as courses, 28,785 students as users and  
284 32,593 registrations as enrolments; login, role and course context come from Moodle, and  
285 cross-role and cross-course requests were refused when tested (Fig. 4). In Canvas, the  
286 dashboard was embedded through an External URL module item, because the available  
287 Canvas image predates LTI 1.3 support, so role resolution on that path relies on the  
288 dashboard’s own authentication. Both integrations establish technical feasibility rather than  
289 deployment readiness.



290  
291 Fig. 4. At-risk insights page of the Moodle plugin in course FFF-2014J, showing the SHAP and LIME  
292 reasoning trail for one student.

## 5. Results

### 5.1. Predictive performance

Table 3 reports out-of-fold performance for the three models. The prototype XGBoost model reached an accuracy of 0.8774, precision of 0.8659, recall of 0.9085, F1-score of 0.8867 and ROC-AUC of 0.9624. Its confusion matrix (Fig. 5) contains 15,634 true positives, 1,574 false negatives, 12,963 true negatives and 2,422 false positives, giving a specificity of 0.8426; about 13% of positive flags were therefore false. Random Forest obtained the highest F1-score (0.8976), accuracy and precision, while XGBoost had the highest recall. The spread in F1-score across the three models was 0.011. Under the recall-first selection criterion, XGBoost was retained for the prototype.

Table 3. Out-of-fold performance under four-fold GroupKFold cross-validation grouped by course presentation (n = 32,593; threshold 0.5).

| Model             | Accuracy      | Precision     | Recall        | F1-score      | Prototype |
|-------------------|---------------|---------------|---------------|---------------|-----------|
| XGBoost           | 0.8774        | 0.8659        | <b>0.9085</b> | 0.8867        | Yes       |
| Random Forest     | <b>0.8921</b> | <b>0.8997</b> | 0.8955        | <b>0.8976</b> | No        |
| Gradient Boosting | 0.8869        | 0.8904        | 0.8962        | 0.8932        | No        |

Note. ROC-AUC for XGBoost was 0.9624. Bold marks the best value in each column.

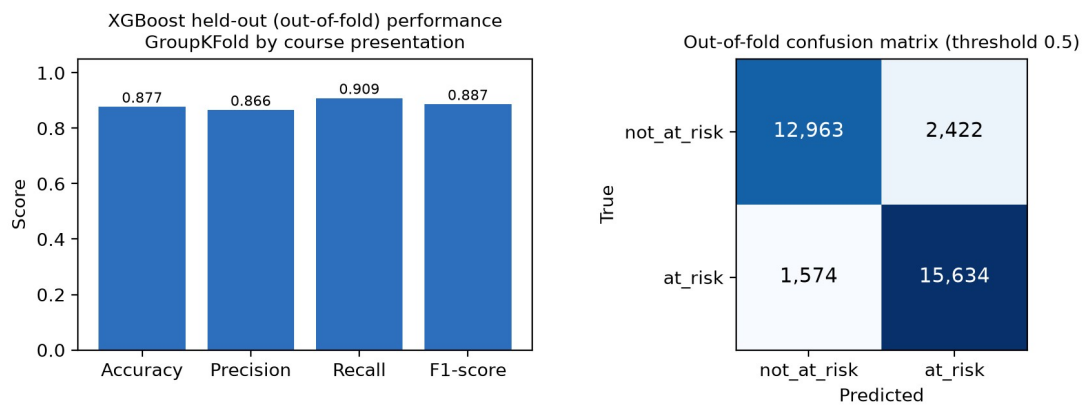


Fig. 5. Out-of-fold accuracy, precision, recall and F1-score of the XGBoost prototype (left) and its confusion matrix at the 0.5 threshold (right), across all 32,593 observations.

### 5.2. Variability across course presentations

Because each fold holds out one presentation, performance can be reported for each unseen presentation (Table 4, Fig. 6). Accuracy ranged from 0.8456 (2014J) to 0.9161 (2013J), a spread of more than seven percentage points, and the error pattern reversed: the 2013

presentations had precision above 0.95 with recall of 0.86 to 0.88, while the 2014 presentations had precision of 0.81 to 0.82 with recall up to 0.9627. With only four presentation-level folds, the source of this variation cannot be attributed precisely, but the model's operating point was not stable across cohorts even with the model and threshold held fixed.

Table 4. XGBoost out-of-fold performance with each course presentation held out in turn.

| Presentation | n      | At-risk rate | Accuracy | Precision | Recall | F1-score |
|--------------|--------|--------------|----------|-----------|--------|----------|
| 2013B        | 4,684  | 0.5527       | 0.9065   | 0.9506    | 0.8764 | 0.9120   |
| 2013J        | 8,845  | 0.4941       | 0.9161   | 0.9690    | 0.8577 | 0.9099   |
| 2014B        | 7,804  | 0.5697       | 0.8619   | 0.8243    | 0.9627 | 0.8882   |
| 2014J        | 11,260 | 0.5154       | 0.8456   | 0.8075    | 0.9197 | 0.8600   |

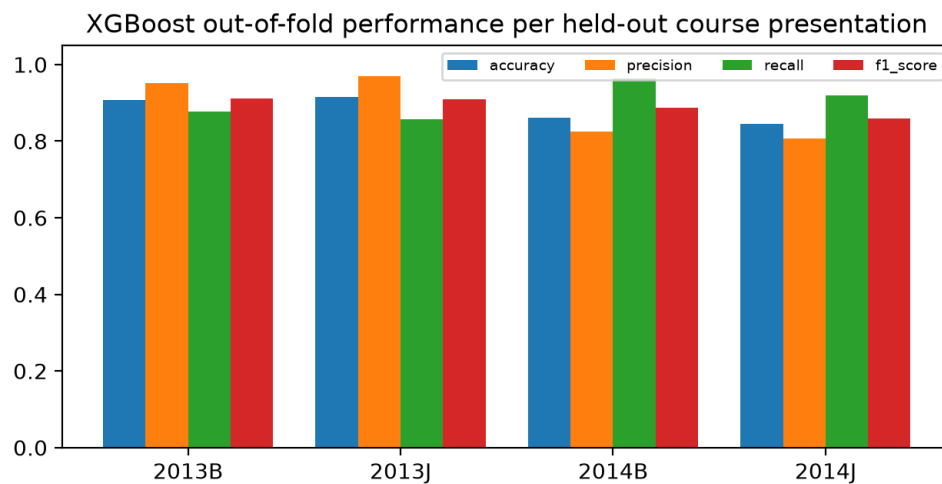


Fig. 6. XGBoost out-of-fold accuracy, precision, recall and F1-score for each held-out course presentation.

### 5.3. Global SHAP explanations

The number of assessments submitted dominated the global TreeSHAP ranking, with a mean absolute SHAP value of 3.1110, more than four times that of the second feature, active days (0.7087) (Table 5, Fig. 7). For the number of assessments submitted, mean score, active days, total clicks and minimum score, the correlation between feature value and SHAP value was strongly negative ( $-0.69$  to  $-0.96$ ), so higher values pushed predictions away from the at-risk class. The 28-day pre-deadline click window was an exception, with a positive correlation ( $+0.8354$ ), in the opposite direction to total clicks and active days. The engagement variables

are highly collinear (Spearman correlation of 0.98 between the 28-day window and total clicks; 0.95 and 0.92 for the 14- and 7-day windows), and SHAP attributions are affected by collinearity (Salih et al., 2025). This is one possible explanation, but no dependence analysis was carried out, so the sign is reported as an observed property of the fitted model and not as evidence that late-window activity is itself a risk factor.

Table 5. Top 10 features by global TreeSHAP importance and by aggregate LIME weight.

| Rank | SHAP feature                   | Mean   SHAP | LIME feature (40 instances)    | Mean   weight |
|------|--------------------------------|-------------|--------------------------------|---------------|
| 1    | Assessments submitted          | 3.1110      | Assessments submitted          | 0.3961        |
| 2    | Active days                    | 0.7087      | Module = FFF                   | 0.1397        |
| 3    | Module = FFF                   | 0.5338      | Module = AAA                   | 0.0995        |
| 4    | Mean score                     | 0.4790      | Mean score                     | 0.0640        |
| 5    | Total clicks                   | 0.3819      | Active days                    | 0.0498        |
| 6    | Clicks 28 days before deadline | 0.3111      | Presentation length            | 0.0447        |
| 7    | Minimum score                  | 0.2741      | Clicks 28 days before deadline | 0.0236        |
| 8    | Presentation length            | 0.2718      | Minimum score                  | 0.0183        |
| 9    | Module = EEE                   | 0.2285      | Registration date              | 0.0073        |
| 10   | Mean days late                 | 0.1468      | Region = Wales                 | 0.0043        |

Note. SHAP values are in log-odds units, computed with the final fitted model over all 32,593 observations. Seven features appear in both top-10 lists. Mean instance-level top-5 overlap: 0.635 (SD 0.197).

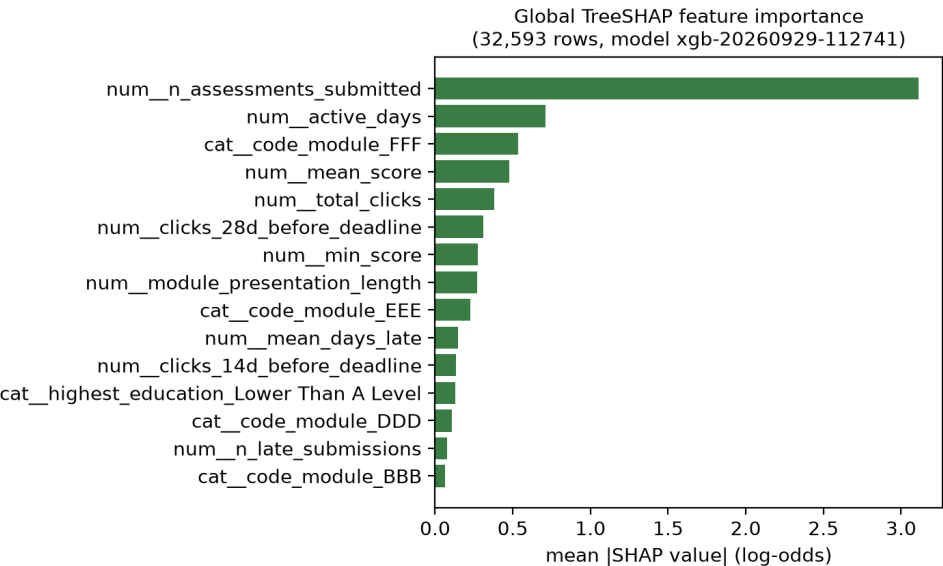


Fig. 7. Top 15 features by global TreeSHAP importance (mean absolute SHAP value), computed with the final fitted model across all 32,593 observations.

#### 5.4. Agreement between SHAP and LIME

Aggregated over the 40 sampled instances, LIME also ranked the number of assessments submitted first (mean absolute weight 0.3961), followed by the FFF and AAA module indicators and mean score (Fig. 8). The two aggregate top-10 rankings shared seven features: assessments submitted, active days, module FFF, mean score, the 28-day click window, minimum score and presentation length. SHAP additionally ranked total clicks, module EEE and mean lateness in its top ten, whereas LIME ranked module AAA, registration date and the Wales region indicator. At the instance level, the mean top-5 overlap was 0.635 (SD 0.197): on average just over three of a student's top five SHAP features also appeared in that student's top five LIME features. LIME tended to rank module indicators higher, consistent with its local, sample-based construction being more sensitive to instances in which module identity is locally decisive. In representative cases, such as a student in FFF-2014J predicted at 0.9998 (true label at risk) and a student in FFF-2014B predicted at 0.0038 (true label not at risk), both methods led with the number of assessments submitted.

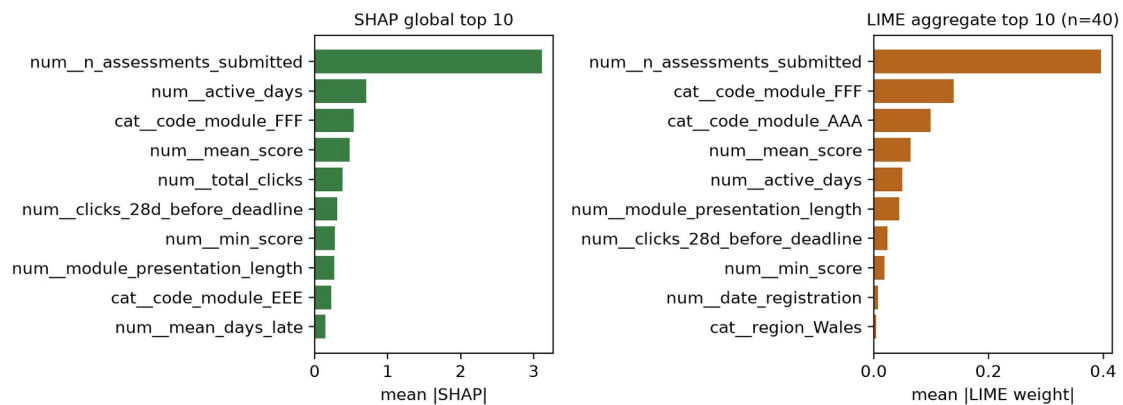


Fig. 8. Global SHAP top-10 features (left) and aggregate LIME top-10 features over 40 sampled instances (right).

#### 5.5. Stability of near-duplicate feature rankings

The 28-day click window (0.3111) and total clicks (0.3819) sat close together in the SHAP ranking. Across eight fits with different random seeds, the 28-day window ranked above total clicks in four fits and below it in the other four, while the combined importance of the two features stayed within 0.57 to 0.69. Their relative order is therefore not a stable property of the model: where two features carry almost the same information, the ensemble distributes

credit between them according to its random subsampling. The combined share of pre-deadline and total engagement was stable and well below that of assessment submission. In a separate execution, the out-of-fold metrics were reproduced exactly, matching those recorded in the model manifest, including the confusion matrix cell for cell.

## **6. Discussion**

### *6.1. Predictive performance under a strict protocol*

The figures in Table 3 are lower than the highest accuracies reported on OULAD, such as the 96.9% reported by Wang (2025), but they are not comparable. Grouping folds by presentation means every prediction was made by a model that had not seen the student's cohort, which is closer to applying an existing model to a new presentation than the stratified random splits used in much of the literature (Jin et al., 2024; Krstajic et al., 2014). The per-presentation results reinforce this: one fixed model under one fixed protocol varied by more than seven accuracy points depending only on which presentation was held out. If protocol and cohort produce this much variation within a single evaluation, numerical differences between studies that differ in split, label definition and feature horizon carry little information about relative model quality (Pelánek, 2018).

The error profile is asymmetric: recall exceeded precision at the untuned 0.5 threshold, so the model leans towards flagging. The 1,574 false negatives are students for whom no role would have received a basis to act, while the 2,422 false positives represent staff effort directed at students who did not withdraw or fail. The comparison with Random Forest shows that the choice of selection metric encodes a judgement about which error matters more, and stating that judgement explicitly is preferable to inheriting it from a general-purpose metric. Because the operating point changed between cohorts, an institution could not fully anticipate the error profile for a new cohort, which argues for periodic revalidation and for showing the current operating point to the roles that rely on it.

### *6.2. What the explanations show*

The concentration of attribution on the number of assessments submitted has a benefit and a cost. Assessment submission is a concrete behaviour that an instructor or advisor can act on,

but a reasoning trail dominated by one feature offers less differentiated information than an explanation layer might suggest, and may be read as confirming what is already visible in the gradebook. The direction of that association is also ambiguous, since a student who has decided to withdraw will stop submitting work. The attributions describe how the fitted model reached its output, not what caused a student's situation, and the dashboard and this paper use the language of contribution and association rather than cause.

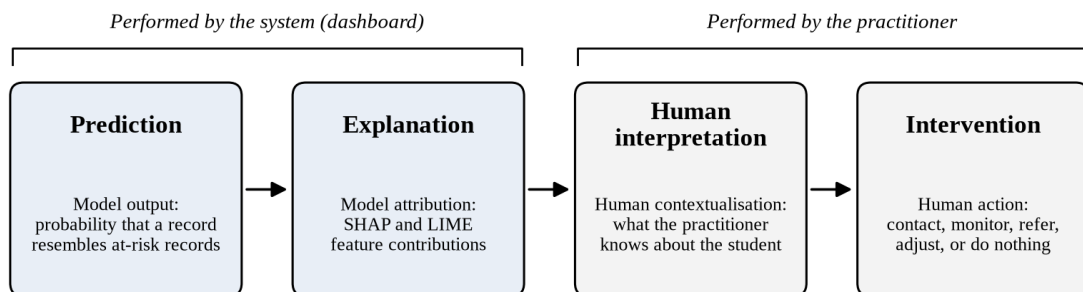
The partial agreement between the methods is the most distinctive explainability finding. The methods agree strongly in aggregate but are not interchangeable for a single prediction, which is the level at which an instructor or advisor would consult them. This is not an error in either method: TreeSHAP computes exact Shapley values for the tree ensemble, whereas LIME fits a sparse local surrogate to sampled perturbations, so the two answer related but different questions (Lundberg et al., 2020; Ribeiro et al., 2016). The measured overlap converts the principled argument for showing both methods (Salih et al., 2025) into an argument from evidence within this system: showing one method alone would hide a measurable degree of disagreement. Where the methods agree, a practitioner has a firmer basis for treating a contributor as a stable part of the model's reasoning; where they diverge, the appropriate response is more caution about that explanation. The finding is consistent with the variability reported by Hooshyar and Yang (2024) and adds a quantity to it for this model and dataset. The 28-day window result shows why direction must be inspected alongside magnitude: a ranking alone would have suggested that heavier late activity is a warning sign, which the evidence does not support.

### *6.3. Role-aware design and the limits of the system*

The rationale for differentiating views is that the three roles act on different objects. The prediction and its attribution are the same for each role, but an instructor decides whether to change something in a running course, an advisor decides how to approach a particular student, and an administrator decides where to direct institutional attention. A single view would require at least two of them to aggregate or disaggregate information themselves. The technical results also indicate what each role specifically needs. For advisors, a caseload

421 drawn from a cohort resembling 2014B (recall 0.9627, precision 0.8243) would be larger and  
 422 contain more students who were not going to withdraw, so a caseload view that does not  
 423 show current precision leaves the advisor unable to weigh each flag. For administrators, the  
 424 between-cohort variability is arguably the most useful single output, since it indicates how  
 425 often the model needs revisiting.

426 CommonKADS contributed traceability from institutional context to interface: the Task  
 427 Model's decomposition into risk assessment, explanation generation, role assignment and  
 428 explanation delivery is mirrored by the five-layer architecture, and each view's granularity  
 429 traces to the Agent Model. The Task Model stops at explanation delivery. Intervention is a  
 430 task for the human agent, and the system is designed to support human interpretation without  
 431 taking over the intervention decision (Fig. 9). An attribution identifying low assessment  
 432 submission narrows where a practitioner should look but does not indicate what to do, since a  
 433 student struggling with the material, one managing illness and one who has decided to leave  
 434 may present the same feature values. A flag with a plausible explanation may still be a false  
 435 positive, as roughly 13% of flags were here, and each would have been accompanied by a  
 436 coherent reasoning trail. Whether side-by-side explanations help practitioners calibrate their  
 437 reliance or make the interface harder to read can only be settled by observation; Brdnik et al.  
 438 (2023) show how perceived trust and satisfaction with SHAP and LIME can be measured in a  
 439 learning analytics setting.



440  
 441 Fig. 9. Relationship between prediction, explanation, human interpretation and intervention. The  
 442 system performs the first two stages; interpretation and intervention remain with the practitioner.

#### 443 6.4. *Candidate design guidelines*

444 The findings suggest six guidelines for explainable learning analytics dashboards, offered as  
445 candidates for empirical validation:

- 446 • Report the agreement between explanation methods where more than one is available,  
447 since a single method presents a more settled picture of local model behaviour than the  
448 attributions support.
- 449 • Let users inspect the direction of each contribution, not only its magnitude, and check  
450 for collinearity artefacts before attributions reach practitioners.
- 451 • Use the language of contribution and association rather than causal or prescriptive  
452 language.
- 453 • Show the current operating point (the balance between missed students and false  
454 alarms) to the roles that rely on it, and revalidate it for each new cohort.
- 455 • Keep the boundary between explanation delivery by the system and intervention by the  
456 human, rather than extending the system to recommend actions its explanation methods  
457 cannot justify.
- 458 • Report subgroup performance before the system is used to direct resources, and enforce  
459 role-based access to per-student risk information at more than one layer.

#### 460 6.5. *Transferability*

461 All results come from one distance-learning institution, where the LMS is the primary site of  
462 study and withdrawal is common, and from presentations in 2013 and 2014. Given that  
463 performance moved by more than seven accuracy points across four presentations of the  
464 same modules, a model trained at one institution should be expected to need retraining and  
465 revalidation elsewhere rather than transfer as a fitted artefact. The CommonKADS design  
466 process, the layered architecture, the cached-global and on-demand-local explanation services  
467 and the reasoning trail rendered at several levels of aggregation do not depend on OULAD  
468 and are the parts most likely to transfer. Transfer would require LMS interaction and  
469 assessment data of sufficient completeness, reliably recorded deadlines, an outcome  
470 definition suited to the institution, roles that correspond to those modelled here and a

governance position on the use of per-student risk predictions. Where teaching is largely in person and the LMS holds only supplementary material, the engagement features that carry much of the signal here would be sparse, and the approach would need to be reconsidered at the level of the Knowledge Model.

#### 6.6. *Limitations*

Five limitations constrain these findings. First, no practitioner used the system, so trust, comprehension, usability and decision quality were not measured, and the role-specific design rests on published accounts rather than elicitation. Second, the features were computed over the full presentation, so the evaluation establishes retrospective classification rather than early-warning performance. Third, calibration of the displayed probabilities, subgroup fairness, explanation stability and fidelity, hyperparameter tuning, class-imbalance handling and trajectory features were planned but not completed, and the LIME analysis rests on 40 instances, with explanations computed in a one-hot feature space that splits categorical attributes across several indicators. Fourth, one registration feature (`withdrew_before_start`) partly encodes the outcome, since all 3,097 observations with this flag are labelled at risk; its mean absolute SHAP value in the final model is 0.00, because the same students have no submitted assessments, but it should be removed in future iterations. Fifth, the LMS integrations were tested in local containers rather than institutional installations.

#### 6.7. *Future work*

Three priorities follow directly from these limitations: an early-warning evaluation with features restricted to the information available at fixed points such as weeks 2, 4, 6 and 8 (Adnan et al., 2021); a calibration study comparing isotonic regression and Platt scaling (Guo et al., 2017; Niculescu-Mizil & Caruana, 2005); and a subgroup fairness assessment across gender, age band, disability and IMD band (Hardt et al., 2016; Mitchell et al., 2019). Beyond these, a study with instructors, advisors and administrators should test how the role-specific views are read and used, drawing on the perceived usefulness and ease of use constructs of TAM, and the approach should be applied to data from other institutions and modes of study.

## 498 **7. Conclusion**

499 This study designed, implemented and technically evaluated a role-aware explainable AI  
500 dashboard that delivers predictions and explanations from one model to instructors, advisors  
501 and administrators at different levels of aggregation. On RQ1, a cached global TreeSHAP  
502 service paired with on-demand local SHAP and LIME ran at full dataset scale and produced  
503 attributions traceable to a versioned model and feature contract. On RQ2, the CommonKADS  
504 models yielded a traceable specification of three role-gated views that share one reasoning  
505 trail at individual, caseload and cohort granularity. On RQ3, the technical preconditions for  
506 supporting practitioner interpretation are in place: the model discriminates well on unseen  
507 presentations, and its explanations are human-readable and cross-checked, with a measured  
508 instance-level SHAP and LIME overlap of 0.635 that supports showing both methods.  
509 Results describe retrospective classification on a single dataset, and whether the dashboard  
510 improves practitioners' trust, understanding or decisions remains to be tested. The prototype  
511 provides a verified technical foundation for that evaluation.

## 512 **CRediT authorship contribution statement**

513 Daniel Larry: Conceptualization, Methodology, Software, Formal analysis, Investigation,  
514 Data curation, Visualization, Writing – original draft. Zainab Shamsiya Usman: Supervision,  
515 Methodology, Writing – review & editing. [Authors to confirm.]

## 516 **Declaration of competing interest**

517 The authors declare that they have no known competing financial interests or personal  
518 relationships that could have appeared to influence the work reported in this paper. [Authors  
519 to confirm.]

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522 commercial, or not-for-profit sectors. [Authors to confirm.]

## 523 **Ethics statement**

524 The study used only the publicly available, anonymised Open University Learning Analytics  
525 Dataset. No primary data were collected from students, instructors, advisors or  
526 administrators.

## 527 **Data availability**

528 OULAD is publicly available (Kuzilek et al., 2017). The prototype source code, including  
529 data preparation, model training, explanation services, the Flask API, the React dashboard  
530 and the Moodle plugin, is available at [https://github.com/daniel-larry/role-aware-xai-](https://github.com/daniel-larry/role-aware-xai-dashboard)  
531 dashboard.

## 532 **Declaration of generative AI and AI-assisted technologies in the writing process**

533 During the preparation of this work the authors used Claude (Anthropic) to condense the  
534 master's thesis on which this article is based into journal format and to verify and format the  
535 references. After using this tool, the authors reviewed and edited the content as needed and  
536 take full responsibility for the content of the published article. [Authors to confirm.]

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538 This article is based on the first author's MSc thesis in Computer Science at the American  
539 University of Nigeria.

## 540 **Appendix A. Reproducibility details**

541 Table A.1. Reproducibility details of the evaluated pipeline.

| Item                | Value                                                                                                                                                              |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Software            | Python 3.11.16; scikit-learn 1.9.1; xgboost 3.2.0; pandas 3.0.6; NumPy 2.4.6; SHAP 0.51.0; LIME 0.2.0.1; joblib 1.6.0; Flask 3.1.3; PyJWT 2.15.1; gunicorn 26.2.0. |
| Hardware            | Docker container on a Linux x86_64 host with 4 virtual CPUs and 16 GB of memory; no GPU.                                                                           |
| Code version        | <a href="https://github.com/daniel-larry/role-aware-xai-dashboard">github.com/daniel-larry/role-aware-xai-dashboard</a> , branch main, commit 7c0068b.             |
| Model artefact      | Serialised joblib pipeline (preprocessing and model) with a JSON manifest; identifier xgb-20260929-112741.                                                         |
| Model configuration | XGBoost, 300 estimators, maximum depth 5, learning rate 0.05, subsample 0.8, column subsample 0.8; threshold 0.5; no resampling or hyperparameter search.          |

| Item         | Value                                                                                                                                      |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Validation   | GroupKFold, four folds grouped by code_presentation (2013B, 2013J, 2014B, 2014J); out-of-fold metrics over all 32,593 observations.        |
| Random seeds | random_state = 42 throughout; LIME sample of 40 instances with NumPy default_rng, seed 42; seed-stability refits with random_state 1 to 7. |

## Appendix B. Feature definitions

Features marked "Full presentation" were computed over the whole course presentation and are not available at an early-warning point as constructed.

Table B.1. Input features of the evaluated model.

| Feature                                                                        | Definition                                                      | Availability      |
|--------------------------------------------------------------------------------|-----------------------------------------------------------------|-------------------|
| num_of_prev_attempts                                                           | Number of previous attempts at the module                       | Registration      |
| studied_credits                                                                | Total credits of the modules being studied                      | Registration      |
| date_registration                                                              | Registration date relative to module start (days)               | Registration      |
| withdrew_before_start                                                          | Unregistered on or before the module start (see Section 6.6)    | Before start      |
| module_presentation_length                                                     | Length of the module presentation (days)                        | Before start      |
| n_assessments_submitted                                                        | Number of assessments submitted                                 | Full presentation |
| mean_score                                                                     | Mean score of submitted assessments                             | Full presentation |
| min_score                                                                      | Lowest score of submitted assessments                           | Full presentation |
| mean_days_late                                                                 | Mean lateness of submissions (on-time counted as 0)             | Full presentation |
| n_late_submissions                                                             | Number of submissions after the deadline                        | Full presentation |
| n_banked                                                                       | Assessment results carried over from a previous presentation    | Full presentation |
| clicks_7d/14d/28d_before_deadline                                              | VLE clicks in the 7, 14 and 28 days before assessment deadlines | Full presentation |
| total_clicks                                                                   | Total VLE clicks                                                | Full presentation |
| active_days                                                                    | Number of days with VLE activity                                | Full presentation |
| mean_daily_clicks                                                              | Total clicks divided by active days                             | Full presentation |
| code_module, gender, region, highest_education, imd_band, age_band, disability | Categorical attributes, one-hot encoded                         | Registration      |

## 549 **References**

- 550 Adnan, M., Habib, A., Ashraf, J., Mussadiq, S., Raza, A. A., Abid, M., Bashir, M., & Khan,  
551 S. U. (2021). Predicting at-risk students at different percentages of course length for  
552 early intervention using machine learning models. *IEEE Access*, 9, 7519–7539.  
553 <https://doi.org/10.1109/ACCESS.2021.3049446>
- 554 Afzaal, M., Nouri, J., Zia, A., Papapetrou, P., Fors, U., Wu, Y., Li, X., & Weegar, R. (2021).  
555 Explainable AI for data-driven feedback and intelligent action recommendations to  
556 support students self-regulation. *Frontiers in Artificial Intelligence*, 4, Article 723447.  
557 <https://doi.org/10.3389/frai.2021.723447>
- 558 Ahmed, W., Wani, M. A., Plawiak, P., Meshoul, S., Mahmoud, A., & Hammad, M. (2025).  
559 Machine learning-based academic performance prediction with explainability for  
560 enhanced decision-making in educational institutions. *Scientific Reports*, 15(1),  
561 Article 26879. <https://doi.org/10.1038/s41598-025-12353-4>
- 562 Ali, S., Abuhmed, T., El-Sappagh, S., Muhammad, K., Alonso-Moral, J. M., Confalonieri, R.,  
563 Guidotti, R., Del Ser, J., Díaz-Rodríguez, N., & Herrera, F. (2023). Explainable  
564 Artificial Intelligence (XAI): What we know and what is left to attain Trustworthy  
565 Artificial Intelligence. *Information Fusion*, 99, Article 101805.  
566 <https://doi.org/10.1016/j.inffus.2023.101805>
- 567 Baker, R. S., & Hawn, A. (2022). Algorithmic bias in education. *International Journal of*  
568 *Artificial Intelligence in Education*, 32(4), 1052–1092.  
569 <https://doi.org/10.1007/s40593-021-00285-9>
- 570 Barredo Arrieta, A., Díaz-Rodríguez, N., Del Ser, J., Bennetot, A., Tabik, S., Barbado, A.,  
571 García, S., Gil-López, S., Molina, D., Benjamins, R., Chatila, R., & Herrera, F.  
572 (2020). Explainable Artificial Intelligence (XAI): Concepts, taxonomies,  
573 opportunities and challenges toward responsible AI. *Information Fusion*, 58, 82–115.  
574 <https://doi.org/10.1016/j.inffus.2019.12.012>
- 575 Brdnik, S., Podgorelec, V., & Šumak, B. (2023). Assessing perceived trust and satisfaction  
576 with multiple explanation techniques in XAI-enhanced learning analytics. *Electronics*,  
577 12(12), 2594. <https://doi.org/10.3390/electronics12122594>

578 Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.  
579 <https://doi.org/10.1023/A:1010933404324>

580 Cabral, L., Pinto, R., & Gonçalves, G. (2025). AI-powered learning analytics dashboards: A  
581 systematic review of applications, techniques, and research gaps. *Discover Education*,  
582 4(1), Article 525. <https://doi.org/10.1007/s44217-025-00964-y>

583 Chavan, P., & Mitra, R. (2022). Tcherly: A teacher-facing dashboard for online video  
584 lectures. *Journal of Learning Analytics*, 9(3), 125–151.  
585 <https://doi.org/10.18608/jla.2022.7555>

586 Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of*  
587 *the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data*  
588 *Mining*, 785–794. <https://doi.org/10.1145/2939672.2939785>

589 Conati, C., Barral, O., Putnam, V., & Rieger, L. (2021). Toward personalized XAI: A case  
590 study in intelligent tutoring systems. *Artificial Intelligence*, 298, Article 103503.  
591 <https://doi.org/10.1016/j.artint.2021.103503>

592 Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of  
593 information technology. *MIS Quarterly*, 13(3), 319–340.  
594 <https://doi.org/10.2307/249008>

595 Ehsan, U., Liao, Q. V., Muller, M., Riedl, M. O., & Weisz, J. D. (2021). Expanding  
596 explainability: Towards social transparency in AI systems. *Proceedings of the 2021*  
597 *CHI Conference on Human Factors in Computing Systems*, 1–19.  
598 <https://doi.org/10.1145/3411764.3445188>

599 Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *The*  
600 *Annals of Statistics*, 29(5), 1189–1232. <https://doi.org/10.1214/aos/1013203451>

601 Guevara-Reyes, R., Ortiz-Garcés, I., Andrade, R., Cox-Riquetti, F., & Villegas-Ch, W.  
602 (2025). Machine learning models for academic performance prediction:  
603 Interpretability and application in educational decision-making. *Frontiers in*  
604 *Education*, 10, Article 1632315. <https://doi.org/10.3389/feduc.2025.1632315>

605 Gunasekara, S., & Saarela, M. (2025). Explainable AI in education: Techniques and  
 606 qualitative assessment. *Applied Sciences*, 15(3), 1239.  
 607 <https://doi.org/10.3390/app15031239>

608 Guo, C., Pleiss, G., Sun, Y., & Weinberger, K. Q. (2017). On calibration of modern neural  
 609 networks. *Proceedings of the 34th International Conference on Machine Learning*,  
 610 *Proceedings of Machine Learning Research*, 70, 1321–1330.  
 611 <https://proceedings.mlr.press/v70/guo17a.html>

612 Gutiérrez, F., Seipp, K., Ochoa, X., Chiluiza, K., De Laet, T., & Verbert, K. (2020). LADA:  
 613 A learning analytics dashboard for academic advising. *Computers in Human*  
 614 *Behavior*, 107, Article 105826. <https://doi.org/10.1016/j.chb.2018.12.004>

615 Haque, A. B., Islam, A. K. M. N., & Mikalef, P. (2023). Explainable artificial intelligence  
 616 (XAI) from a user perspective: A synthesis of prior literature and problematizing  
 617 avenues for future research. *Technological Forecasting and Social Change*, 186,  
 618 Article 122120. <https://doi.org/10.1016/j.techfore.2022.122120>

619 Hardt, M., Price, E., & Srebro, N. (2016). Equality of opportunity in supervised learning.  
 620 *Advances in Neural Information Processing Systems*, 29, 3315–3323.  
 621 <https://proceedings.neurips.cc/paper/2016/hash/6a9659feb1216f14f7384ba499518b38>  
 622 -Abstract.html

623 Hevner, A. R., March, S. T., Park, J., & Ram, S. (2004). Design science in information  
 624 systems research. *MIS Quarterly*, 28(1), 75–106. <https://doi.org/10.2307/25148625>

625 Hlosta, M., Zdrahal, Z., & Zendulka, J. (2017). Ouroboros: Early identification of at-risk  
 626 students without models based on legacy data. *Proceedings of the Seventh*  
 627 *International Learning Analytics & Knowledge Conference*, 6–15.  
 628 <https://doi.org/10.1145/3027385.3027449>

629 Hooshyar, D., & Yang, Y. (2024). Problems with SHAP and LIME in interpretable AI for  
 630 education: A comparative study of post-hoc explanations and neural-symbolic rule  
 631 extraction. *IEEE Access*, 12, 137472–137490.  
 632 <https://doi.org/10.1109/ACCESS.2024.3463948>

633 Hussain, M., Zhu, W., Zhang, W., & Abidi, S. M. R. (2018). Student engagement predictions  
634 in an e-learning system and their impact on student course assessment scores.  
635 *Computational Intelligence and Neuroscience*, 2018, Article 6347186.  
636 <https://doi.org/10.1155/2018/6347186>

637 Jin, L., Wang, Y., Song, H., & So, H.-J. (2024). Predictive modelling with the Open  
638 University Learning Analytics Dataset (OULAD): A systematic literature review. In  
639 A. M. Olney, I.-A. Chounta, Z. Liu, O. C. Santos, & I. I. Bittencourt (Eds.), *Artificial*  
640 *intelligence in education. Posters and late breaking results, workshops and tutorials,*  
641 *industry and innovation tracks, practitioners, doctoral consortium and blue sky* (pp.  
642 477–484). Springer. [https://doi.org/10.1007/978-3-031-64315-6\\_46](https://doi.org/10.1007/978-3-031-64315-6_46)

643 Jones, M., Bradley, J., & Sakimura, N. (2015). *JSON Web Token (JWT)* [RFC 7519]. Internet  
644 Engineering Task Force. <https://doi.org/10.17487/RFC7519>

645 Kaliisa, R., & Dolonen, J. A. (2023). CADA: A teacher-facing learning analytics dashboard  
646 to foster teachers' awareness of students' participation and discourse patterns in  
647 online discussions. *Technology, Knowledge and Learning*, 28(3), 937–958.  
648 <https://doi.org/10.1007/s10758-022-09598-7>

649 Kaliisa, R., Jivet, I., & Prinsloo, P. (2023). A checklist to guide the planning, designing,  
650 implementation, and evaluation of learning analytics dashboards. *International*  
651 *Journal of Educational Technology in Higher Education*, 20(1), Article 28.  
652 <https://doi.org/10.1186/s41239-023-00394-6>

653 Khosravi, H., Buckingham Shum, S., Chen, G., Conati, C., Tsai, Y.-S., Kay, J., Knight, S.,  
654 Martinez-Maldonado, R., Sadiq, S., & Gašević, D. (2022). Explainable artificial  
655 intelligence in education. *Computers and Education: Artificial Intelligence*, 3, Article  
656 100074. <https://doi.org/10.1016/j.caeai.2022.100074>

657 Kostopoulos, G., Davrazos, G., & Kotsiantis, S. (2024). Explainable artificial intelligence-  
658 based decision support systems: A recent review. *Electronics*, 13(14), 2842.  
659 <https://doi.org/10.3390/electronics13142842>

660 Krstajic, D., Buturovic, L. J., Leahy, D. E., & Thomas, S. (2014). Cross-validation pitfalls  
 661 when selecting and assessing regression and classification models. *Journal of*  
 662 *Cheminformatics*, 6(1), Article 10. <https://doi.org/10.1186/1758-2946-6-10>  
 663 Kuzilek, J., Hlosta, M., & Zdrahal, Z. (2017). Open University Learning Analytics dataset.  
 664 *Scientific Data*, 4(1), Article 170171. <https://doi.org/10.1038/sdata.2017.171>  
 665 Leitner, P., Ebner, M., & Ebner, M. (2019). Learning analytics challenges to overcome in  
 666 higher education institutions. In D. Ifenthaler, D.-K. Mah, & J. Y.-K. Yau (Eds.),  
 667 *Utilizing learning analytics to support study success* (pp. 91–104). Springer.  
 668 [https://doi.org/10.1007/978-3-319-64792-0\\_6](https://doi.org/10.1007/978-3-319-64792-0_6)  
 669 Liao, Q. V., & Varshney, K. R. (2022). *Human-centered explainable AI (XAI): From*  
 670 *algorithms to user experiences* (arXiv:2110.10790). arXiv.  
 671 <https://doi.org/10.48550/arXiv.2110.10790>  
 672 Lundberg, S. M., Erion, G., Chen, H., DeGrave, A., Prutkin, J. M., Nair, B., Katz, R.,  
 673 Himmelfarb, J., Bansal, N., & Lee, S.-I. (2020). From local explanations to global  
 674 understanding with explainable AI for trees. *Nature Machine Intelligence*, 2(1), 56–  
 675 67. <https://doi.org/10.1038/s42256-019-0138-9>  
 676 Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions.  
 677 *Advances in Neural Information Processing Systems*, 30, 4765–4774.  
 678 <https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b6776>  
 679 7-Abstract.html  
 680 Matcha, W., Uzir, N. A., Gašević, D., & Pardo, A. (2020). A systematic review of empirical  
 681 studies on learning analytics dashboards: A self-regulated learning perspective. *IEEE*  
 682 *Transactions on Learning Technologies*, 13(2), 226–245.  
 683 <https://doi.org/10.1109/TLT.2019.2916802>  
 684 Miller, T. (2019). Explanation in artificial intelligence: Insights from the social sciences.  
 685 *Artificial Intelligence*, 267, 1–38. <https://doi.org/10.1016/j.artint.2018.07.007>  
 686 Mitchell, M., Wu, S., Zaldivar, A., Barnes, P., Vasserman, L., Hutchinson, B., Spitzer, E.,  
 687 Raji, I. D., & Gebru, T. (2019). Model cards for model reporting. *Proceedings of the*

688 *Conference on Fairness, Accountability, and Transparency*, 220–229.  
689 <https://doi.org/10.1145/3287560.3287596>

690 Mohseni, S., Zarei, N., & Ragan, E. D. (2021). A multidisciplinary survey and framework for  
691 design and evaluation of explainable AI systems. *ACM Transactions on Interactive*  
692 *Intelligent Systems*, 11(3–4), Article 24. <https://doi.org/10.1145/3387166>

693 Molenaar, I., & Knoop-van Campen, C. A. N. (2019). How teachers make dashboard  
694 information actionable. *IEEE Transactions on Learning Technologies*, 12(3), 347–  
695 355. <https://doi.org/10.1109/TLT.2018.2851585>

696 Niculescu-Mizil, A., & Caruana, R. (2005). Predicting good probabilities with supervised  
697 learning. *Proceedings of the 22nd International Conference on Machine Learning*,  
698 625–632. <https://doi.org/10.1145/1102351.1102430>

699 Nnadi, L. C., Watanobe, Y., Rahman, Md. M., & John-Otumu, A. M. (2024). Prediction of  
700 students’ adaptability using explainable AI in educational machine learning models.  
701 *Applied Sciences*, 14(12), 5141. <https://doi.org/10.3390/app14125141>

702 Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M.,  
703 Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D.,  
704 Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in  
705 Python. *Journal of Machine Learning Research*, 12, 2825–2830.  
706 <https://jmlr.org/papers/v12/pedregosa11a.html>

707 Peffers, K., Tuunanen, T., Rothenberger, M. A., & Chatterjee, S. (2007). A design science  
708 research methodology for information systems research. *Journal of Management*  
709 *Information Systems*, 24(3), 45–77. <https://doi.org/10.2753/MIS0742-1222240302>

710 Pelánek, R. (2018). The details matter: Methodological nuances in the evaluation of student  
711 models. *User Modeling and User-Adapted Interaction*, 28(3), 207–235.  
712 <https://doi.org/10.1007/s11257-018-9204-y>

713 Poursabzi-Sangdeh, F., Goldstein, D. G., Hofman, J. M., Wortman Vaughan, J., & Wallach,  
714 H. (2021). Manipulating and measuring model interpretability. *Proceedings of the*  
715 *2021 CHI Conference on Human Factors in Computing Systems*, Article 237.  
716 <https://doi.org/10.1145/3411764.3445315>

717 Rebelo Marcolino, M., Reis Porto, T., Thompsen Primo, T., Targino, R., Ramos, V., Marques  
 718 Queiroga, E., Munoz, R., & Cechinel, C. (2025). Student dropout prediction through  
 719 machine learning optimization: Insights from Moodle log data. *Scientific Reports*,  
 720 15(1), Article 9840. <https://doi.org/10.1038/s41598-025-93918-1>  
 721 Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). “Why should I trust you?”: Explaining the  
 722 predictions of any classifier. *Proceedings of the 22nd ACM SIGKDD International*  
 723 *Conference on Knowledge Discovery and Data Mining*, 1135–1144.  
 724 <https://doi.org/10.1145/2939672.2939778>  
 725 Rudin, C. (2019). Stop explaining black box machine learning models for high stakes  
 726 decisions and use interpretable models instead. *Nature Machine Intelligence*, 1(5),  
 727 206–215. <https://doi.org/10.1038/s42256-019-0048-x>  
 728 Salih, A. M., Raisi-Estabragh, Z., Boscolo Galazzo, I., Radeva, P., Petersen, S. E., Lekadir,  
 729 K., & Menegaz, G. (2025). A perspective on explainable artificial intelligence  
 730 methods: SHAP and LIME. *Advanced Intelligent Systems*, 7(1), Article 2400304.  
 731 <https://doi.org/10.1002/aisy.202400304>  
 732 Saltzer, J. H., & Schroeder, M. D. (1975). The protection of information in computer  
 733 systems. *Proceedings of the IEEE*, 63(9), 1278–1308.  
 734 <https://doi.org/10.1109/PROC.1975.9939>  
 735 Schreiber, G., Akkermans, H., Anjewierden, A., de Hoog, R., Shadbolt, N. R., Van de Velde,  
 736 W., & Wielinga, B. J. (2000). *Knowledge engineering and management: The*  
 737 *CommonKADS methodology*. MIT Press.  
 738 <https://doi.org/10.7551/mitpress/4073.001.0001>  
 739 Schwendimann, B. A., Rodríguez-Triana, M. J., Vozniuk, A., Prieto, L. P., Boroujeni, M. S.,  
 740 Holzer, A., Gillet, D., & Dillenbourg, P. (2017). Perceiving learning at a glance: A  
 741 systematic literature review of learning dashboard research. *IEEE Transactions on*  
 742 *Learning Technologies*, 10(1), 30–41. <https://doi.org/10.1109/TLT.2016.2599522>  
 743 Siemens, G., & Long, P. (2011). Penetrating the fog: Analytics in learning and education.  
 744 *EDUCAUSE Review*, 46(5), 30–40.

745 [https://er.educause.edu/articles/2011/9/penetrating-the-fog-analytics-in-learning-and-](https://er.educause.edu/articles/2011/9/penetrating-the-fog-analytics-in-learning-and-education)  
746 [education](https://er.educause.edu/articles/2011/9/penetrating-the-fog-analytics-in-learning-and-education)

747 Slack, D., Hilgard, S., Jia, E., Singh, S., & Lakkaraju, H. (2020). Fooling LIME and SHAP:  
748 Adversarial attacks on post hoc explanation methods. *Proceedings of the AAAI/ACM*  
749 *Conference on AI, Ethics, and Society*, 180–186.  
750 <https://doi.org/10.1145/3375627.3375830>

751 Sokol, K., & Flach, P. (2020). Explainability fact sheets: A framework for systematic  
752 assessment of explainable approaches. *Proceedings of the 2020 Conference on*  
753 *Fairness, Accountability, and Transparency*, 56–67.  
754 <https://doi.org/10.1145/3351095.3372870>

755 Springer, A., & Whittaker, S. (2020). Progressive disclosure: When, why, and how do users  
756 want algorithmic transparency information? *ACM Transactions on Interactive*  
757 *Intelligent Systems*, 10(4), 1–32. <https://doi.org/10.1145/3374218>

758 Studer, R., Benjamins, V. R., & Fensel, D. (1998). Knowledge engineering: Principles and  
759 methods. *Data & Knowledge Engineering*, 25(1–2), 161–197.  
760 [https://doi.org/10.1016/S0169-023X\(97\)00056-6](https://doi.org/10.1016/S0169-023X(97)00056-6)

761 Susnjak, T., Ramaswami, G. S., & Mathrani, A. (2022). Learning analytics dashboard: A tool  
762 for providing actionable insights to learners. *International Journal of Educational*  
763 *Technology in Higher Education*, 19(1), Article 12. [https://doi.org/10.1186/s41239-](https://doi.org/10.1186/s41239-021-00313-7)  
764 [021-00313-7](https://doi.org/10.1186/s41239-021-00313-7)

765 Tsai, Y.-S., & Gašević, D. (2017). Learning analytics in higher education: Challenges and  
766 policies. *Proceedings of the Seventh International Learning Analytics & Knowledge*  
767 *Conference*, 233–242. <https://doi.org/10.1145/3027385.3027400>

768 Verbert, K., Duval, E., Klerkx, J., Govaerts, S., & Santos, J. L. (2013). Learning analytics  
769 dashboard applications. *American Behavioral Scientist*, 57(10), 1500–1509.  
770 <https://doi.org/10.1177/0002764213479363>

771 Visani, G., Bagli, E., Chesani, F., Poluzzi, A., & Capuzzo, D. (2022). Statistical stability  
772 indices for LIME: Obtaining reliable explanations for machine learning models.

773         *Journal of the Operational Research Society*, 73(1), 91–101.  
774         <https://doi.org/10.1080/01605682.2020.1865846>  
775     Wang, Y. (2025). Artificial intelligence in student management systems to enhance academic  
776         performance monitoring and intervention. *Scientific Reports*, 15(1), Article 35122.  
777         <https://doi.org/10.1038/s41598-025-19159-4>  
778     Yu, R., Li, Q., Fischer, C., Doroudi, S., & Xu, D. (2020). Towards accurate and fair  
779         prediction of college success: Evaluating different sources of student data. In A. N.  
780         Rafferty, J. Whitehill, V. Cavalli-Sforza, & C. Romero (Eds.), *Proceedings of the*  
781         *13th International Conference on Educational Data Mining* (pp. 292–301).  
782         International Educational Data Mining Society.